1. Import Libraries ¶

In [1]:
```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
```

2. Load the Data

In [2]:
```
data = pd.read_excel("Infrastructure.xlsx")
```

3. Data Information and Initial Cleaning

In [3]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20874 entries, 0 to 20873
Data columns (total 4 columns):
 #   Column                                  Non-Null Count  Dtype  
---  ------                                  --------------  -----  
 0   Inund Depth obs. (m)                    20874 non-null  float64
 1   Damage Indices (0-4)                    20874 non-null  int64  
 2   Material (RC, Steel, Wood, Masonry)     20874 non-null  object 
 3   Structure (Building or Infrastructure)  20874 non-null  object 
dtypes: float64(1), int64(1), object(2)
memory usage: 652.4+ KB
```

In [4]:
```
data.drop(columns=["Structure (Building or Infrastructure)"], inplace=True)
```

In [5]:
```
data.head()
```

Out[5]:

	Inund Depth obs. (m)	Damage Indices (0-4)	Material (RC, Steel, Wood, Masonry)
0	2.1	2	Masonry
1	1.8	2	Masonry
2	2.2	2	Masonry
3	2.0	2	Wood
4	2.0	2	Wood

4.Renaming Columns for Clarity

In [6]:
```
data.rename(columns={
    'Inund Depth obs. (m)': 'Inund Depth',
    'Damage Indices (0-4)': 'Damage Indices',
    'Material (RC, Steel, Wood, Masonry)': 'Material'
}, inplace=True)

# Print the DataFrame with renamed columns
data.head()
```

Out[6]:

	Inund Depth	Damage Indices	Material
0	2.1	2	Masonry
1	1.8	2	Masonry
2	2.2	2	Masonry
3	2.0	2	Wood
4	2.0	2	Wood

In [7]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20874 entries, 0 to 20873
Data columns (total 3 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   Inund Depth     20874 non-null  float64
 1   Damage Indices  20874 non-null  int64  
 2   Material        20874 non-null  object 
dtypes: float64(1), int64(1), object(1)
memory usage: 489.4+ KB
```

In [8]:
```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
data['Inund Depth'] = scaler.fit_transform(data[['Inund Depth']])  # Original c
```

5. Value Counts and Filtering the Data

In [9]:
```
data['Inund Depth'].value_counts()
```

Out[9]:
```
0.081197    615
0.076923    602
0.068376    554
0.085470    542
0.072650    537
           ...
0.786325      1
0.055983      1
0.017094      1
0.112393      1
0.053419      1
Name: Inund Depth, Length: 296, dtype: int64
```

In [10]:
```
data["Damage Indices"].value_counts()
```

Out[10]:
```
2    7155
4    6891
3    5221
1    1428
0     179
Name: Damage Indices, dtype: int64
```

In [11]:
```
data["Damage Indices"].value_counts()
```

Out[11]:
```
2    7155
4    6891
3    5221
1    1428
0     179
Name: Damage Indices, dtype: int64
```

In [12]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20874 entries, 0 to 20873
Data columns (total 3 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Inund Depth     20874 non-null  float64
 1   Damage Indices  20874 non-null  int64
 2   Material        20874 non-null  object
dtypes: float64(1), int64(1), object(1)
memory usage: 489.4+ KB
```

In [13]:
```
data.drop(data[(data['Damage Indices'] == 0)].index, inplace=True)
```

In [14]:
```
data["Damage Indices"].value_counts()
```

Out[14]:
```
2    7155
4    6891
3    5221
1    1428
Name: Damage Indices, dtype: int64
```

In [15]:
```
data.shape
```

Out[15]:
```
(20695, 3)
```

In [16]:
```
data.duplicated().sum()
```

Out[16]:
```
19374
```

In [17]:
```
data['Inund Depth'].value_counts()
```

Out[17]:
```
0.081197    606
0.076923    594
0.068376    549
0.085470    534
0.072650    531
           ...
0.126068      1
0.076068      1
0.057692      1
0.096581      1
0.053419      1
Name: Inund Depth, Length: 286, dtype: int64
```

6. One-Hot Encoding for Categorical Variables

In [18]:
```
pd.get_dummies(data,columns=["Damage Indices","Material"])
```

Out[18]:

	Inund Depth	Damage Indices_1	Damage Indices_2	Damage Indices_3	Damage Indices_4	Material_Masonry	Material_Reinforced concrete
0	0.128205	0	1	0	0	1	0
1	0.115385	0	1	0	0	1	0
2	0.132479	0	1	0	0	1	0
3	0.123932	0	1	0	0	0	0
4	0.123932	0	1	0	0	0	0
...	...	...	...	...	...	...	...
20869	0.376068	0	0	0	1	0	0
20870	0.376068	0	0	0	1	0	0
20871	0.141026	0	0	1	0	0	0
20872	0.354701	0	0	0	1	0	0
20873	0.397436	0	0	1	0	0	0

20695 rows × 9 columns

In [19]:
```
data = pd.get_dummies(data,columns=["Damage Indices","Material"],drop_first= Fa
data.head()
```

Out[19]:

	Inund Depth	Damage Indices_1	Damage Indices_2	Damage Indices_3	Damage Indices_4	Material_Masonry	Material_Reinforced concrete	M
0	0.128205	0	1	0	0	1	0	
1	0.115385	0	1	0	0	1	0	
2	0.132479	0	1	0	0	1	0	
3	0.123932	0	1	0	0	0	0	
4	0.123932	0	1	0	0	0	0	

In [20]:
```
data.shape
```

Out[20]: (20695, 9)

In [21]:
```
data.duplicated().sum()
```

Out[21]: 19374

7. Define Features and Target Variables

In [22]:
```
# Define features (X) and target (y)
X = data[['Inund Depth', 'Material_Reinforced concrete', 'Material_Steel', 'Mat
y = data[['Damage Indices_1','Damage Indices_2', 'Damage Indices_3', 'Damage Ir
```

In [23]:
```
y = np.argmax(y.values, axis=1)
```

8. Split the Data into Training, Validation, and Test Sets

In [24]:
```
from sklearn.model_selection import train_test_split

# First, split the data into 90% training+validation and 10% testing
X_train_val, X_test, y_train_val, y_test = train_test_split(X, y, test_size=0.2

# Further split the training+validation set into training (80%) and validation
X_train, X_val, y_train, y_val = train_test_split(X_train_val, y_train_val, tes
```

9. Initialize and Train the Random Forest Model

In [25]:
```
# Step 4: Initialize and Train the Random Forest Model
rf = RandomForestClassifier(n_estimators=100, random_state=0)
rf.fit(X_train, y_train)
```

Out[25]:
```
RandomForestClassifier
(https://scikit-learn.org/1.6/modules/generated/sklearn.ensemble.Ra

RandomForestClassifier(random_state=0)
```

10. Evaluate the Model on the Test Set

In [26]:
```
# Step 6: Evaluate on the Test Set
y_test_pred = rf.predict(X_test)
test_accuracy = accuracy_score(y_test, y_test_pred)
print("Test Accuracy:", test_accuracy)
```

```
Test Accuracy: 0.7612949987919787
```

11. Classification Report

In [27]:
```
# Step 7: Detailed Classification Report
print("Classification Report (Test Set):")
print(classification_report(y_test, y_test_pred))
```

```
Classification Report (Test Set):
              precision    recall  f1-score   support

           0       0.77      0.25      0.38       289
           1       0.79      0.88      0.83      1469
           2       0.64      0.69      0.66      1037
           3       0.83      0.80      0.81      1344

    accuracy                           0.76      4139
   macro avg       0.76      0.65      0.67      4139
weighted avg       0.76      0.76      0.75      4139
```

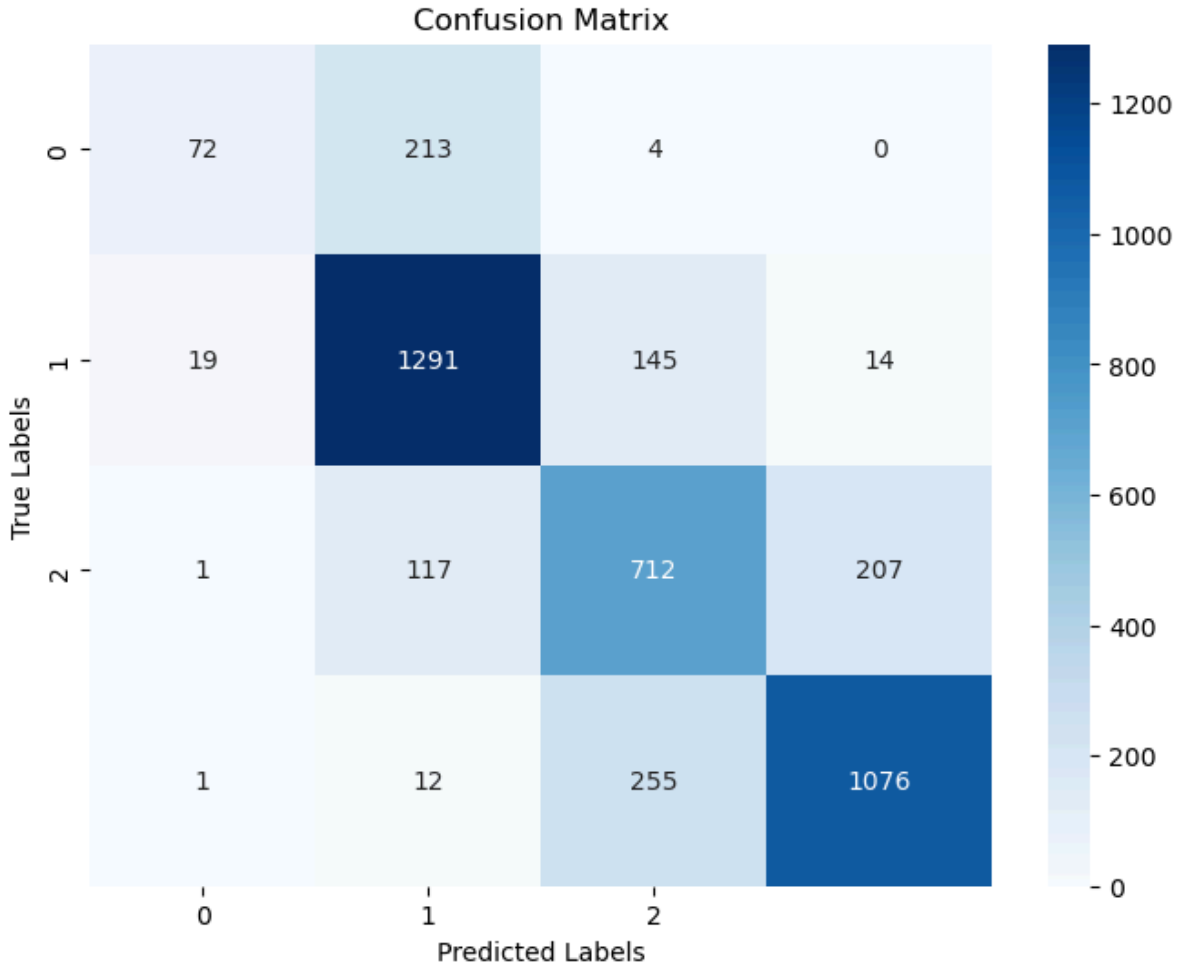
In [28]:
```
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_test_pred)


plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=[0, 1, 2], ytick


plt.title("Confusion Matrix")
plt.xlabel("Predicted Labels")
plt.ylabel("True Labels")


plt.show()
```



In [ ]:

In [ ]: